MILITARY SPECIFICATION

GYROSCOPE, RATE INTEGRATING

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers the general requirements for rate integrating gyroscopes for use in inertial navigation, shipboard and airborne fire control system.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids shall form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Cadmium Plating (Electrodeposited)
PPP-B-621	Boxes, Wood, Nailed and Lock-corner

MILITARY

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-16910	Interference Measurement, Methods and Limits
MIL-M-3171A	Magnesium Alloy, Corrosion Protection
MIL-M-45202	Magnesium Alloy, Anodic Treatment of
MIL-P-116	Preservation, Methods of
MIL-Q-9858	Quality Control System Requirement

FSC 6615

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MIL-S-6872 Soldering Processes, General Specifications for
MIL-W-16878/4 Wire, Electrical, Type E, 200°C

STANDARDS

MILITARY

MIL-STD-129	Marking of Shipments
MIL-STD-143	Specifications and Standards, Order of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MS-33586	Metals, Definition of Dissimilar

DRAWINGS

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions shall be obtained from the procuring activity.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28 Screw-Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Detail specifications for individual gyro types. All requirements specified herein are applicable to all types of integrating gyros covered by military specifications. Detail requirements or exceptions applicable to individual types will be specified in the detail gyro specifications. In the event of any conflict between

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requirements of this specification and requirements of the detail gyro specification, the latter shall govern.

3.2 Definitions and Conventions.

3.2.1 Rate integrating gyro. A rate integrating gyro is a single degree of freedom gyro having primarily viscous restraint of its spin axis about the output axis. In this gyro an output signal is produced by gimbal angular displacement, relative to the case, which is proportional to the integral of the angular rate of the case about the input axis. The term gyro is synonymous with gyroscope for this specification.

3.2.2 Rotor moment of inertia. Rotor moment of inertia is the moment of inertia of a gyro rotor about its spin axis.

3.2.3 Spin axis (SA). The spin axis is the axis of rotation of the rotor.

3.2.4 Spin reference axis (SRA). The spin reference axis is the nominal direction of the spin axis, indicated by external case marking, when the gyro output has specified values, usually null.

3.2.5 Degree of freedom. A degree of freedom is an allowable mode of angular motion of the spin axis with respect to the case. The number of degrees of freedom is the number of orthogonal axes about which the spin axis is free to rotate.

3.2.6 Axis of freedom. An axis of freedom is an axis about which a gimbal provides a degree of freedom.

3.2.7 Gimbal. A gimbal is a support which provides the spin axis with a degree of freedom.

3.2.8 Input axis (IA). An input axis is an axis about which a rotation of the case causes a maximum output. It is normal to the spin axis.

3.2.9 Input reference axis. An input reference axis is the nominal direction of an input axis as indicated by external case markings. It is nominally normal to the spin reference axis.

3.2.10 Output axis (OA). An output axis is an axis of freedom provided with a pickoff which generates an output signal as a function of the output angle.

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3.2.11 Precession. Precession is a rotation of the spin axis produced by a torque, T, applied about an axis mutually perpendicular to the spin axis and the axis of the resulting rotation. A constant precession rate, W, is related to rotor angular momentum, H, and applied torque, T, by the equation WH = T.

3.2.12 Command rate. Command rate is the input rate equivalent to a precession rate caused by a torquer command signal.

3.2.13 Capturing. Capturing is the use of a torquer to restrain one or more gyro axes or gimbal axes to a specified reference position.

3.2.14 Slewing. Slewing is the rotation of the spin axis about an axis parallel to that of the applied torque causing the rotation.

3.2.15 Earth's rate correction. Earth's rate correction is a command rate applied to a gyro to compensate for the drift rate of the gyro spin axis with respect to its case caused by the rotation of the earth.

3.2.16 Pickoff. A pickoff is a device which produces a signal output, generally a voltage as a function of the relative angular displacement of two elements. This angular displacement is usually between two gimbals or between a gimbal and the case.

3.2.17 Torquer. A torquer is a device which exerts torque on a gimbal in response to an input signal.

3.2.18 Cross coupling errors. Cross coupling errors are errors in the gyro output resulting from gyro sensitivity to inputs about axes normal to an input reference axis.

3.2.19 Electrical zero. Electrical zero is the angular reference position of a pickoff where, with a certain excitation applied to the pickoff, its in-phase fundamental output will be zero.

3.2.20 Current product (MA^2). The in-phase product of the current flowing in two windings of a microsyn. (i.e., $I_1 I_2 \cos \phi = MA^2$).

Where I_1 = Reference winding current

I_2 = Control winding current

ϕ = Phase angle of I_1 to I_2

Positive current product is the product of reference and control winding currents of the same sign and produces positive torque.

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3.2.21 Scale factor. Scale factor is the ratio of a change in output to a change in the input intended to be measured. Scale factor is generally evaluated as the slope of a straight line that can be fitted by the method of least squares to input - output data obtained by varying the input cyclically over the input range. Scale factors of interest for the gyros include; torquer scale factor (dyne-cm per milliamp), pickoff scale factor (volts per degree output angle), command rate scale factor (degrees per hour per milliamp), and gyro scale factor (volts per degree input angle).

3.2.22 Output range. Output range is the product of input range and scale factor.

3.2.23 Input limits. Input limits are the extreme values of the input, generally plus and minus, within which performance is of the specified quality. Input limits are generally expressed either as angular displacements or angular rate, or both.

3.2.24 Input range. Input range is the absolute value of the difference between the input limits.

3.2.25 Linearity error. Linearity error is a measure of the greatest variation of the input output data from the least squares straight line. It is the maximum output deviation from the least squares straight line expressed as a percent of one-half the output range.

3.2.26 Hysteresis. Hysteresis is the difference between output signals for increasing and decreasing inputs at that input for which the difference is maximum, measured after cycling through the input range.

3.2.27 Composite error. Composite error is the maximum deviation of the output data from the specified output function. Composite error is due to the composite effects of hysteresis, resolution, non-linearity, repeatability, and other uncertainties in the output data. It is expressed as a percentage of one-half the output range.

3.2.28 Error band. Error band is a band about the specified output function which contains the output data. The error band contains the composite effects of hysteresis, resolution, non-linearity, repeatability and other uncertainties in the output data.

3.2.29 Drift rate. Drift rate is the time rate of angular deviation of the spin axis from the desired orientation in space. It consists of random and systematic drift rate components. It is expressed as an equivalent input angular rate in degrees per unit time.

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3.2.30 Acceleration insensitive drift rate. Acceleration insensitive drift rate consists of those components of systematic drift rate that have no correlation with acceleration. It is ascribable to such effects as temperature gradients and reaction torques. It is expressed in degrees per unit time.

3.2.31 Acceleration sensitive drift rate. Acceleration sensitive drift rate consists of those components of systematic drift rate that are correlated with translational acceleration applied to the gyro case. The relationship of these components of drift rate to acceleration can be stated by means of coefficients having units such as degrees per unit time per "g" (e.g. mass unbalance) and degrees per unit time per "g" squared (e.g. drift caused by anisoelasticity). The acceleration squared coefficient relates the drift rate to the product of the components of the instantaneous value of the acceleration resolved along the principle axes of the gyro.

3.2.32 Mass unbalance. Mass unbalance is the characteristic of the gyro resulting from lack of coincidence of the center of suspension and the center of mass. It gives rise to torques caused by translational accelerations which lead to acceleration sensitive drift rates.

3.2.33 Anisoelasticity. Anisoelasticity is the inequality of compliance of the rotor and gimbal assembly in different directions. When the gyro is subjected to undirectional or vibratory accelerations at an angle to a principal axis of compliance anisoelasticity leads to torques proportional to the square of the applied accelerations. This torque results in acceleration sensitive drift rate proportional to acceleration squared.

3.2.34 Reaction torques. Reaction torques are torques exerted on a gimbal as a result of electrical excitations applied to the gyro exclusive of torquer command signals. This torque usually results in an acceleration insensitive drift rate.

3.2.35 Sensitivity. Sensitivity is the ratio of a change in output to a change in an undesirable or secondary input. For example, the drift rate temperature sensitivity of a gyro is the ratio of the change in drift rate to a change in temperature.

3.2.36 Viscous restraint. The viscous restraint coefficient (C) refers to the viscous restraint of the flotation fluid in dyne-cm/radian/second.

3.3 Selection of government documents. Specification and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

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3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Parts, materials, processes and workmanship. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. A detailed description of the materials, design and manufacturing processes must be submitted as part of the quality assurance procedure in compliance with MIL-Q-9858.

3.4.1 Critical materials. Non-critical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.4.2 Metals. Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the gyroscope may be subjected when in storage or during normal service life. Unless suitably protected against electrolytic corrosion dissimilar metals, as defined in MS-33586 shall not be used in intimate contact with each other.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the gyroscope except where magnetic materials are essential.

3.4.4 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

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3.4.5 Corrosive fumes. The materials as installed in the gyroscope and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.6 Finish. Protective coatings, potting compounds and finishes which will, crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.4.7 Aluminum parts. Where practical, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Aluminum alloys which do not anodize satisfactorily, shall be coated with a chemical film in accordance with MIL-C-5541.

3.4.8 Magnesium parts. Where practical, magnesium alloy parts shall be plated in accordance with MIL-M-3171A or MIL-M-45202 and of a class that is adequate to achieve the necessary degree of protection.

3.4.9 Steel parts. Where practical, steel parts shall be plated in accordance with QQ- N-416, QQ-N-290 or QQ-C-320, and of a class that is adequate to achieve the degree of protection required.

3.4.10 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872.

3.4.11 Workmanship. The gyro, including all parts and accessories, shall be contructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine screw assemblies and freedom of parts from burrs and sharp edges.

3.4.12 Cleaning. The gyro shall be thoroughly cleaned and loose spattered or excess solder, flux, metal chips and other foreign material shall be removed.

3.4.13 Wire. Electrical wire used for terminal leads shall conform to MIL-W-16878/4 (Navy).

3.5 Dielectric strength. Gyroscopes shall display no evidence of insulation breakdown or internal arcing when tested in accordance with 4.13 at the test points specified in the detail specification.

3.6 Insulation resistance. Insulation resistance measurements shall be performed in accordance with 4.14. The resistance shall be at least 100 megohms.

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3.7 Spin motor.

3.7.1 Type. The spin motor shall be the type specified in the detail specification.

3.7.2 Winding configuration. The spin motor winding configuration shall be as specified in the detail specification.

3.7.3 Rated excitation. The spin motor excitation voltage and frequency shall be as specified in the applicable detail specification.

3.7.4 Test excitation. The spin motor test excitation voltage and frequency shall be as specified in the applicable detail specification.

3.7.5 Angular velocity. The spin motor angular velocity when running at synchronous speed shall be as specified in the applicable detail specification.

3.7.6 Rotor angular momentum. Rotor angular momentum, H, is the product of spin angular velocity and rotor moment of inertia. It is a measure of the ability of a gyro rotor to maintain the spin axis fixed in space. The angular momentum shall be as specified in the detail specification. Angular momentum is expressed in gram centimeters squared per second.

3.7.7 Run-up time. Run-up time is the time interval required for the gyro rotor to reach a specified speed from stand still. When tested in accordance with 4.15.1, the run-up time shall be within the limits specified in the applicable detail specification.

3.7.8 Power consumption. When tested in accordance with 4.15.2, the power consumption of the spin motor shall not exceed the value specified in the detail specification.

3.7.9 Starting current. The starting current shall not exceed the value specified in the detail specification when tested in accordance with 4.15.3.

3.7.10 Running current. The spin motor running current shall be within the value specified in the detail specification when tested in accordance with 4.15.4.

3.8 Pickoff.

3.8.1 Type. The pickoff shall be the type specified in the detail specification.

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3.8.2 Rated excitation. The pickoff excitation shall be as specified in the applicable detail specification.

3.8.3 Test excitation. The pickoff test excitation shall be as specified in the detail specification.

3.8.4 Impedance. When tested in accordance with 4.16, the impedance of the pickoff windings shall be within the limits specified in the detail specification.

3.8.5 Null voltage. When tested in accordance with 4.17.1, the null voltage shall not exceed the value specified in the detail specification.

3.8.6 Phase shift. Phase shift is the difference between the time phase of the fundamental component of the primary voltage and the time phase of the secondary voltage of the pickoff with the gimbal at the positive stop. When tested in accordance with 4.17.2, the value of phase shift shall be as specified in the detail specification.

3.8.7 Linearity error. When tested in accordance with 4.17.3, the linearity error of the angle input to voltage output of the pickoff shall be as specified in the detail specification.

3.9 Torquer.

3.9.1 Type. The torquer shall be the type specified in the applicable detail specification.

3.9.2 Reference winding excitation. The maximum reference winding excitation shall be as specified in the detail specification.

3.9.3 Control winding excitation. The maximum control winding excitation shall be as specified in the detail specification.

3.9.4 Test excitation. The torquer test excitation of the reference winding shall be as specified in the detail specification.

3.9.5 Command rate scale factor. The command rate scale factor is the ratio of output angular velocity to torquer input. Command rate scale factor is evaluated as the slope of the least squares straight line of input-output data obtained by varying the input cyclically over the input range. When tested in accordance with 4.18.1, the command rate scale factor shall be as specified in the detail specification.

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3.9.6 Linearity error. When tested in accordance with 4.18.1, the maximum torquer linearity error shall be as specified in the detail specification. Linearity error is defined in 3.2.25.

3.9.7 Hysteresis. When tested in accordance with 4.18.1, the hysteresis shall not exceed the value specified in the detail specification. Hysteresis is defined in 3.2.26.

3.9.8 Maximum command rate. The maximum command rate shall be as specified in the detail specification.

3.9.9 Impedance. When tested in accordance with 4.16, the impedance of the torquer windings shall be as specified in the detail specification.

3.9.10 Null voltage. When tested in accordance with 4.18.2, the null voltage shall not exceed the value specified in the detail specification.

3.9.11 Reaction torque. Torquer reaction torque is the unwanted torque produced on a gimbal as a result of electric current in the reference winding. When tested in accordance with 4.18.3, the torquer reaction torque shall not exceed the limit specified in the detail specification.

3.10 Temperature sensor.

3.10.1 Type. The temperature sensor shall be of the type specified in the applicable detail specification .

3.10.2 Scale factor. The temperature scale factor is the change in resistance per degree fahrenheit. The sign of the scale factor is positive when the resistance increases with temperature. The scale factor of the temperature sensor shall be as specified in the detail specification when tested in accordance with 4.19.1.

3.10.3 Operating temperature resistance. The operating temperature resistance shall be as specified in the detail specification. See 4.10.3 stabilized operating temperature.

3.10.4 Induced voltage. When tested in accordance with 4.19.2, the induced voltage shall not exceed the value specified in the detail specification.

3.10.5 Test excitation. The temperature sensor test excitation shall be as specified in the detail specification.

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3.10.6 Operating temperature. The operating temperature is that temperature at which the temperature sensing element shall stabilize at the nominal value of operating temperature resistance specified in 3.10.3. When tested in accordance with 4.19.3, the operating temperature shall be within the limits specified in the detail specification.

3.10.7 Over temperature sensor. The over temperature sensor shall be as specified in the detail specification.

3.11 Heaters.

3.11.1 Power. When tested in accordance with 4.20, the power consumed by the heaters at specified operating voltage shall be within the limits specified in the applicable detail specification.

3.11.2 Test excitation. The standby heater shall be excited with the type of power specified in the detail specification. The gyro shall be maintained at operating temperature by maintaining the temperature sensing winding at operating temperature resistance as specified in 3.10.3. The temperature sensing winding is connected to a temperature control device which regulates the gyro temperature by controlling the power to the standby heater.

3.12 Gyro scale factor. The gyro scale factor ($Eo/\emptyset i$) is the ratio of resultant pickoff secondary voltage (Eo) to a gyro input axis displacement ($\emptyset$). When tested in accordance with 4.21, the gyro scale factor shall be as specified in the detail specification.

3.13 Input angular freedom. The input angular freedom is the amount of input axis angular rotation required to precess the gimbal against its mechanical stop. Input angular freedom is measured for positive and negative input angles. When tested in accordance with 4.22 the input angular freedom shall be as specified in the detail specification.

3.14 Gyro phasing. Gyro axes and positive rotation are defined in Figure 1. For positive wheel rotation about the spin axis, a positive rotation about the input axis produces positive rotation of the gimbal about the output axis. Positive torque from the torquer also causes a positive output about the output axis. When tested in accordance with 4.23, the gyro phasing shall be as specified in Figure 1 when the spin motor, pickoff and torquer are excited and phased as specified in the detail specification.

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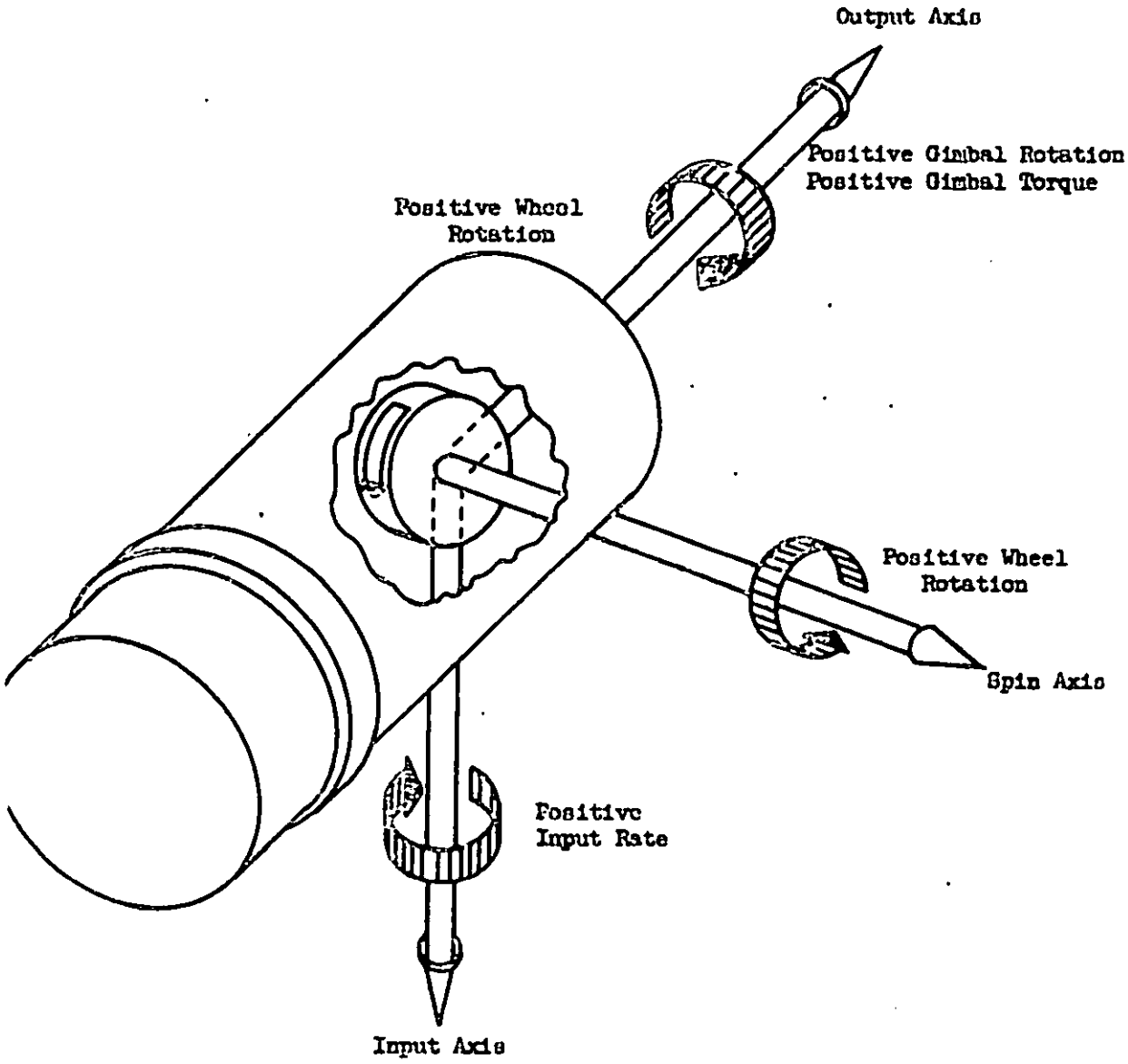


Figure 1 GYRO PHASING DIAGRAM

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3.15 Null repeatability. The null repeatability test gives an indication of friction and constancy of reaction torques. When tested in accordance with 4.24, the null repeatability shall not exceed the value specified in the detail specification.

3.16 Characteristic time. Characteristic time is the time required for the output to reach 63 per cent of its final value for a step input. When tested in accordance with 4.25, the characteristic time shall not exceed the value as specified in the detail specification.

3.17 Axis alignment. The input axis misalignment about the output axis and the spin reference axis shall not exceed the value specified in the detail specification when tested in accordance with 4.26.

3.18 Acceleration insensitive drift rate. When tested in accordance with the method specified in the detail specification the acceleration insensitive drift rate shall conform to the requirements in the detail specification. Drift rate test methods are described in 4.27 through 4.30 of this document.

3.19 Acceleration sensitive drift rate.

3.19.1 Mass unbalance. When tested in accordance with the method specified in the detail specification, drift rate correlated to the first power of linear acceleration shall not exceed the value specified in the detail specification. Mass unbalance test methods are described in 4.27 through 4.29 of this document.

3.19.2 Anisoelasticity. When tested in accordance with the method specified in the detail specification, drift rate correlated to the second power of linear acceleration shall not exceed the value specified in the detail specification. Test methods for measuring anisoelasticity are described in 4.29 and 4.31 of this document.

3.20 Random drift rate. Random drift rate is the time varying component of drift rate under specified operating conditions. It is expressed in degrees per unit time. When tested in accordance with the method specified in the detail specification, the random drift rate shall not exceed the value specified in the applicable detail specification. Test methods for measuring random drift are described in 4.29 and 4.30 of this document.

3.21 Elastic restraint coefficient. Elastic restraint coefficient is the ratio of gimbal restraining torque about the output axis to output angle expressed in degrees per hour per degree. When tested in accordance with 4.32, the elastic restraint coefficient shall not exceed the value specified in the detail specification.

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3.22 Day to day stability. The change of the average value of drift rate in the same gyro orientation measured on three successive days, shall not exceed the value specified in the detail specification when tested in accordance with 4.33.

3.23 Environmental sensitivity. The sensitivity of a gyro to variations in environmental conditions of air currents, magnetic fields, thermal radiation, ambient change, ambient pressure, spin motor excitation variation, spin motor excitation frequency variation, pickoff excitation variation, and torquer excitation variation shall not exceed the values specified in the detail specification. Testing shall be in accordance with 4.34.

3.24 Vibration. The gyroscope shall be capable of withstanding harmonic vibrations at frequencies up to 2000 cycles per second at a force of 15 g's without damage or loosening of parts. After testing in accordance with 4.35, the gyroscope shall meet the requirements of 3.5, 3.6, 3.15, 3.18, 3.19, and 3.20 as applicable.

3.25 Shock. The gyro shall be capable of withstanding shock impacts at an acceleration of 50g and 11 ±1 millisecond duration without damage, loosening of parts or degradation of performance characteristics. After testing in accordance with 4.36, the gyro shall meet the requirements of 3.5, 3.6, 3.15, 3.18, 3.19, and 3.20 as applicable.

3.26 Altitude. The gyro shall be capable of operation from sea level to 100,000 feet. When tested in accordance with 4.37 the gyro shall meet the requirements of 3.6.

3.27 Moisture resistance. Gyros shall be capable of operation and storage in climates of high relative humidity. After testing in accordance with 4.38, gyros shall meet the requirements of 3.5 and 3.6.

3.28 Salt atmosphere resistance. This test is intended to determine whether or not salt laden air will damage the exterior of the gyro case by corrosion, or short circuit to the terminals, etc. After testing in accordance with 4.39, the gyro shall be cleaned with distilled water and then dried. There shall be no visible evidence of corrosion or other damage on the gyro exterior. The gyro shall then meet the requirements of 3.5 and 3.6.

3.29 Ambient temperature. When required by detail specification gyros shall be capable of storage at the ambient temperature range specified in the detail specification. When tested in accordance with 4.40, gyros shall meet the requirements of 3.5, 3.6, 3.15, 3.18, 3.19, and 3.20.

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3.30 Radio noise. When tested in accordance with 4.41, radio interference radiated into space shall be less than 2.5 peak microvolts per meter per kilocycle bandwidth from 0.15 to 65 megacycles and less than 5.0 peak microvolts per meter per kilocycles bandwidth from 55 to 150 megacycles. Radio interference conducted into the line shall not be in excess of the limits in MIL-I-16910 and applicable detail specifications.

3.31 Operating life. Gyros shall be capable of operation for the period specified in the detail specification without failure. After testing in accordance with 4.42, gyros shall meet the requirements specified in 3.5, 3.6, 3.7, 3.8, 3.9, 3.12, 3.15, 3.18, 3.19, and 3.20.

3.32 Nameplate. The nameplate shall be of the size, location and contain the nameplate data as specified in the detail specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified, quality control system requirements are per MIL-Q-9858. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the rate integrating gyro shall be classified as follows: qualification; preproduction; periodic production; and acceptance inspection.

4.3 Qualification. A qualification sample of four rate integrating gyros manufactured in accordance with the requirements of the detail specification shall be subjected to the qualification tests detailed in Schedule 1 of Table I at an activity designated by the procuring activity. Production of the rate integrating gyro prior to approval of the qualification sample, shall be at the supplier's risk. Each rate integrating gyro submitted shall be plainly identified by securely attached durable tags marked with the following information;

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TABLE I SCHEDULE OF TESTS

Schedule 1 No. 1 - No. 40
Schedule 2 No. 1 - No. 29
Schedule 3 Accelerated

	Test		Requirement	Test Method
1.	Visual and Mechanical Inspection	o		4.8
2.	Dielectric Strength	o	3.5	4.13
3.	Insulation Resistance	o	3.6	4.14
4.	Scale Factor		3.10.2	4.19.1
5.	Operating Temperature		3.10.6	4.19.3
6.	Run-up Time	o	3.7.7	4.15.1
7.	Power Consumption		3.7.8	4.15.2
8.	Running Current		3.7.10	4.15.4
9.	Starting Current		3.7.9	4.15.3
10.	Impedance	o	3.8.4	4.16
11.	Impedance	*	3.9.9	4.16
12.	Null Voltage	o	3.8.5	4.17.1
13.	Phase Shift		3.8.6	4.17.2
14.	Pickoff Linearity Error		3.8.7	4.17.3
15.	Null Voltage	*	3.9.10	4.18.2
16.	Induced Voltage		3.10.4	4.19.2
17.	Gyro Scale Factor	o	3.12	4.21
18.	Input Angular Freedom	*	3.13	4.22
19.	Gyro Phasing	o	3.14	4.23
20.	Null Repeatability	*	3.15	4.24
21.	Command Rate Scale Factor	*	3.9.5	4.18.1
22.	Torquer Linearity Error	*	3.9.6	4.18.1
23.	Hysteresis	*	3.9.7	4.18.1
24.	Axis Alignment	*	3.17	4.26
25.	Reaction Torque		3.9.11	4.18.3
26.	Acceleration Insensitive Drift	o	3.18	See note
27.	Acceleration Sensitive Drift	o	3.19	See note
28.	Random Drift	o	3.20	See note
29.	Day to Day Stability		3.22	4.33
30.	Characteristic Time		3.16	4.25
31.	Elastic Restraint		3.21	4.32
32.	Environmental Sensitivity		3.23	4.34
33.	Vibration		3.24	4.35
34.	Shock		3.25	4.36
35.	Altitude		3.26	4.37
36.	Moisture Resistance		3.28	4.39
37.	Salt Atmosphere Resistance		3.28	4.39
38.	Ambient Temperature		3.29	4.40
39.	Radio Noise		3.30	4.41
40.	Operating Life		3.31	4.42

NOTE: Test methods may be found in 4.27 through 4.31. The specific method will be in the detail specification.

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SAMPLE FOR QUALIFICATION INSPECTION

Gyroscope, Rate Integrating (type number, if any)
Submitted by (manufacturer's name, date) for
Qualification tests in accordance with MIL-G- (WP)
Under authorization (reference letter authorizing tests)

4.3.1 Qualification Inspection. Qualification tests shall be performed on rate integrating gyros that are representative of the equipment to be supplied under the contract. The qualification sample that has satisfactorily completed the tests detailed in Schedule 3 of Table I shall be submitted as required by 4.3 for the performance of the qualification tests described in Schedule 1 of Table I of this specification. The manufacturer shall submit certified test results showing satisfactory completion by the sample of all the Schedule 3 Table I tests. Failure of any sample to satisfactorily pass any of the tests shall require the manufacturer to follow the procedure described in 4.8.

4.3.2 Changes. After samples have been submitted for qualification, no change shall be made in the materials, design or manufacturing processes without approval of the qualifying activity.

4.3.2 Qualification approval. Satisfactory completion of the qualification inspection will be sufficient evidence for qualification approval and will place the supplier's name on the Qualified Products List for a period of two years. Re-evaluation of qualification approval will take place every six months, or when any of the following conditions occur:

a. The supplier has modified the gyro design.

b. The supplier has instituted a change in the material used, or in the processing (see 4.3.2).

c. The specification requirements for the gyro have been amended or revised to the extent that the qualifying activity believes that re-evaluation is necessary.

4.3.4 Disposition of sample units. Sample units subjected to qualification tests shall not be delivered on any contract or order.

4.4 Requalification. Qualification tests in accordance with 4.3 shall be repeated every two years, or more frequently, as directed by the procuring or qualifying activity.

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4.5 Preproduction inspection. When requested by the procuring agency, sample gyros taken from the first production lot of each contract shall be submitted for testing to the qualification tests of 4.3.

4.6 Periodic production inspection. Periodic production inspection samples shall be selected from units that have satisfactorily passed the quality conformance inspections of Schedule 3 as follows:

Step 1 - Three rate gyros selected at random from the first lot.

Step 2 - One rate gyro selected at random from each succeeding lot, unless,

a. A major change in design occurs, at which time, required by the procuring activity, samples shall be submitted for preproduction inspection of the approved new design.

b. A unit fails to pass the tests. When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(1) determine the cause of the defect

(2) take the necessary actions to correct or eliminate the defect from all units on hand, and, if so directed by the procuring activity, from those units previously shipped.

(3) submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

The samples shall be forwarded to the testing activity and be subjected to all the tests described in Schedule 2 of this specification. Samples submitted shall be plainly identified by a securely attached durable tag marked with the following information:

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SAMPLE FOR PRODUCTION INSPECTION

Gyroscope, Rate Integrating (type, number, serial number)
submitted by (manufacturer's name, date) for production testing
in accordance with MIL-G- (WP)
Under contract number

4.7 Quality conformance inspection. Each production rate integrating gyro to be delivered under the contract shall be subjected to thequality conformance tests of Schedule 3. Failure to meet any of the requirements therein shall be cause for rejection.

4.8 Production lot size. The lot size for the purpose of sampling tests shall be 100 rate gyros or any fraction thereof remaining on the contract.

4.9 Failure. A gyro failure is any malfunction which causes the gyro to fail to meet the minimum performance requirements specified herein. Failure of any gyro in any test (except insulation tests) shall be cause for withholding approval. Whenever the rate of insulation failures of gyros in a production run (100 units) exceeds two per cent the entire run shall be held for review by the procuring or qualifying activity. When a defective unit is found the contractor shall:

a. determine the cause of the failure.

b. take the necessary actions to correct or eliminate the defect from all units on hand.

c. submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

4.10 Test conditions.

4.10.1 Standard test conditions. Unless otherwise specified herein, all measurements and tests shall be made at an ambient temperature of 72 ± 5°F., atmospheric pressure of 30 ± 2 inches Hg., and a relative humidity no greater than 55 per cent. Rate of change of temperature shall not exceed 5°F. per hour.

4.10.2 Test excitations. Spin motor excitation shall be as specified in 3.7.4. Pickoff excitation shall be as specified in 3.8.3. Torquer excitation shall be as specified in 3.9.4. Temperature sensor excitation shall be as specified in 3.11.2.

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4.10.3 Stabilized operating temperature. The stabilized operating temperature is the temperature of the gyro when the temperature sensing element resistance is within the limits specified in 3.10.3 and has stabilized at that temperature for a period of one hour.

4.10.4 Gyro mounting accuracy. Unless specified otherwise, the test unit should be mounted in such a way that the alignment of its three reference axes with the test fixture is maintained within the limits specified in the detail specification under all specified test conditions. The mounting fixture should be designed such that no undue stress can be exerted on the mounted gyro. The three reference axes of the test unit are those defined by external case markings or mounting surfaces.

4.11 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the required tests. Generally, test equipment accuracy should be ten times greater than the measured tolerance. The contractor shall establish adequate calibration of test equipment to the satisfaction of the procuring activity or approved testing laboratory. The function tests performed fall in one of two categories generally:

a. Servo turntable loop. Tests on a single axis turntable where the gyro output signal controls a servo rotating the turntable and with it the gyro case about its input axis, and

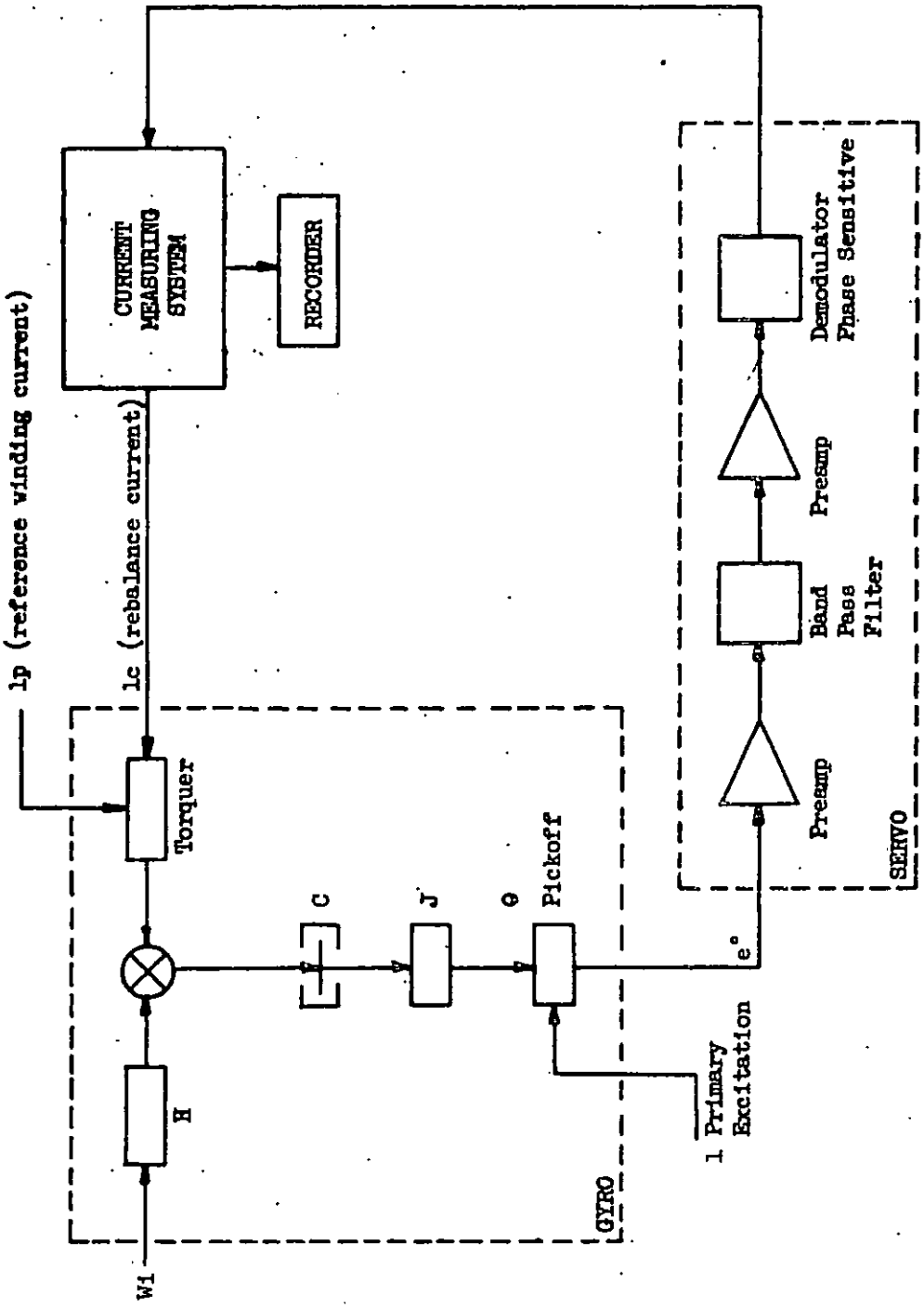
b. Servo rate loop. Tests in which the gyro output signal controls the gyro torquer directly such as to maintain the output axis at null (see Figure 2 and Figure 3).

4.11.1 General precautions and recommendations. In both types of function tests the gyro should be mounted on a test stand sufficiently stable and isolated from disturbing torques and forces in the surrounding structure or building. The selection of test equipment and the design of servo loops should be based on accuracy requirements compatible with the performance specifications. Similarly, the bandpass of the meters used should be chosen such as to provide information within the frequency spectrum of interest for the tests, especially in the case of drift tests. Since most rate integrating gyros are high precision units, special care is required to maintain proper operating conditions and to eliminate external disturbing influences such as temperature gradients and fluctuations. Magnetic fields, stress on the unit due to mounting conditions, etc. All tests should be performed in a well controlled laboratory environment.

4.11.2 Description of test equipment. In addition to a stable base structurally independent of the adjacent floor and a gyro mounting fixture, each of the following is required in one or more of the tests specified:

a. Single-axis servo driven turntable which can be tilted from horizontal to vertical and which is equipped with a precision angular readout.

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SERVO LOOP BLOCK DIAGRAM - FOR D-C TORQUER

Figure 2

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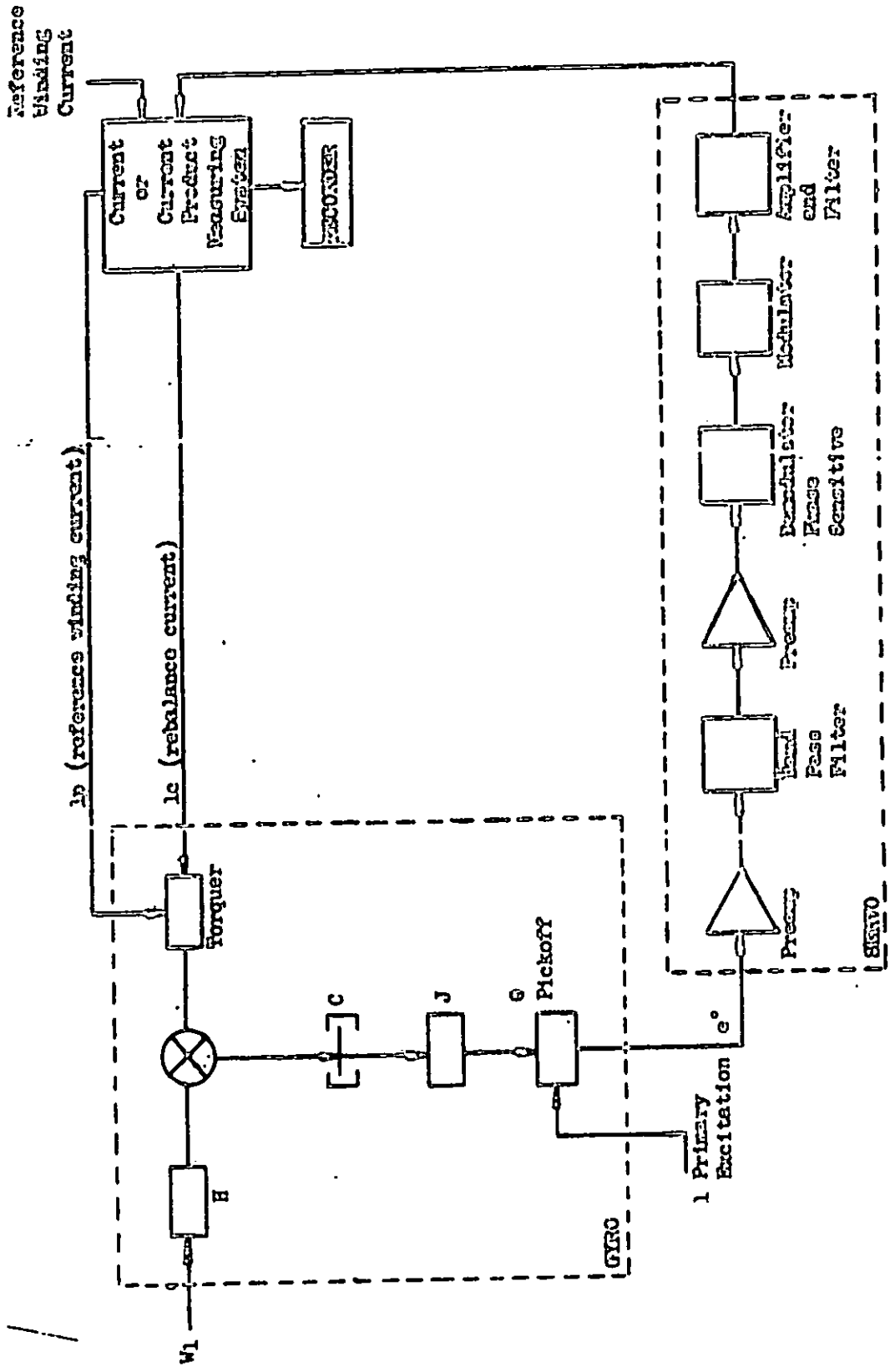


Figure 3

SERVO LOOP BLOCK DIAGRAM - FOR A-C SYSTEM

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b. Single-axis turntable with means of applying and measuring angular displacement inputs (such as dividing head).

c. Single-axis turntable with precision rate drive.

d. Single-axis oscillating table with means of driving it sinusoidally at known variable frequencies.

e. Servo rate loop. The servo rate loop is the servo loop formed by passing the pickoff signal output through and amplifier and phase sensitive demodulator into the torquer control field with the gyro normally excited (see Figure 2). For a - c torquers the system requires that the d - c demodulator signal be modulated and filtered to provide an a - c sine wave rebalance current to the torquer (See Figure 3). The servo gain shall be as specified in the detail specification. The current measuring system normally consists of a precision fixed resistor in series with the torquer control winding, the rebalance current is then proportional to the voltage drop across the resistor. For a - c torquers a current product resolver may be used to obtain a direct measurement of current product. A - c torquers require that the output impedance of the servo be at least 25,000 ohms. Unless otherwise specified the input impedance load of the servo amplifier on the pickoff shall not be less than 0.5 megohm shunted by not more than 350 picofarads.

4.12 Visual and mechanical inspection. The gyro shall be carefully inspected to determine that the dimensions and finishes conform to the applicable detail specification.

4.13 Dielectric strength. Dielectric strength test. A test potential of the magnitude specified by 3.5 shall be applied to the indicated test points. The test potential shall be raised slowly (minimum time 3 seconds) to the specified value and held at that value for one minute; then reduced slowly (minimum time 3 seconds) to zero. The peak value of the test potential through out this test shall not exceed 1.5 times the specified test potential. The gyro shall be maintained at operating temperature as specified in 4.10.3. A failure shall be indicated by a winding leakage current in excess of one peak milliampere or the indication of internal arcing or corona discharge. Immediately after successfully completing the dielectric strength test the insulation resistance test (4.14) shall be performed.

4.14 Insulation resistance test. The d -c resistance between specified test points shall be measured with 500 volts d - c applied. The temperature of the gyro shall be the operating temperature as specified in 4.10.3.

4.15 Spin motor tests.

4.15.1 Run up time. With the gyroscope at operating temperature, the spin motor shall be excited as specified in 3.7.4. The run up time required for the spin motor to reach synchronous speed shall be determined by observing a drop in line current and shall not exceed the value specified in 3.7.7.

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4.15.2 Power consumption. The spin motor power consumption shall be measured with the gyroscope energized and stabilized as specified in 4.10.2 and 4.10.3 and shall not exceed the value specified in 3.7.8.

4.15.3 Starting current. The gyroscope shall be energized and stabilized as specified in 4.10.2 and 4.10.3. Remove the excitation to the spin motor and allow the motor to stop. The spin motor excitation is reapplied and current in each phase measured during run-up time from zero to synchronous speed. The peak current of any phase shall not exceed the value specified in 3.7.9.

4.15.4 Running current. The spin motor running current shall be measured with the gyroscope energized and stabilized as specified in 4.10.2 and 4.10.3. Running current shall not exceed the value specified in 3.7.10.

4.16 Impedance. The gyroscope shall be stabilized as specified in 4.10.3. The inductance and d - c resistance of the pickoff and torquer winding shall be within the limits specified in 3.8.4 and 3.9.9. Measure the inductance using an excitation of 2 milliamperes RMS at 1000 cps.

4.17 Pickoff tests.

4.17.1 Null voltage. The gyroscope shall be mounted in a dividing head and test stand as specified in 4.11 with the input reference axis parallel to the axis of rotation of the dividing head, within ±1.5 degrees. Excite and stabilize the gyro as specified in 4.10.2 and 4.10.3. The null voltage is the minimum pickoff voltage obtained by rotating the dividing head. The voltage measuring devices used shall indicate the average value of the voltage wave in terms of the RMS value of an equivalent sine wave and shall have an input impedance of at least 1 megohm resistance shunted by capacitance of 30 picofarads.

4.17.2 Phase shift. The gyroscope shall be mounted, energized, and stabilized as in 4.17.1 above. Rotate the gimbal to the positive limit by applying a torquer current or rotation about the IA. The phase shift between the pickoff primary and secondary voltages shall be measured using a device which indicates or compares the phase shift to an accuracy of ±15 minutes and does not alter the open circuit voltages by more than 0.1 per cent.

4.17.3 Pickoff linearity error. The gyroscope shall be mounted, energized, and stabilized as in 4.17.1. Apply the input angular displacement specified in the detail specification to the gyro. Measure the corresponding gyro pickoff output. The time for each reading shall not exceed 1 second in order to minimize the effect

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of earth's rate and systematic drift rate. Find the maximum deviation of the input-output data from the least squares straight line. Linearity error in per cent equals the value of this deviation times 100, divided by one-half the input range.

4.18 Torquer tests.

4.18.1 Command rate scale factor, linearity error and hysteresis. Test equipment includes base mounting fixture, excitation supplies, single-axis servo driven turntable, associated servo amplifier and electronics, precision current supply and timer. Mount the test unit with the input reference axis vertical. Energize and stabilize as specified in 4.10.2 and 4.10.3. Connect the gyro pickoff output to the turntable servo to maintain the gyro pickoff output within 5 millivolts of null. Apply the input currents specified in the detail specification to the gyro torquer. Measure the corresponding rate outputs of the servo driven turntable for each of the input currents in the following sequence. Start with positive input limit, return to zero, increase to negative input limit, return to zero, and increase to positive input limit. Repeat this cycle 3 times. In some applications variations of performance characteristics with output angle are of interest. In such cases, the output angles and input currents will be specified for the test series. In order to operate the gyro at an output angle other than zero, a properly phased constant bias voltage shall be inserted in series with the gyro pickoff output. The operational performance characteristics listed below are based on the terms 3.2.25 and 3.2.26. Command rate scale factor is the slope of the straight line that can be fitted by the method of least squares to the input-output data obtained from the test above.

Torquer linearity error. Find the maximum deviation of the input-output data from the least squares straight line. Linearity error in per cent equals the value of this deviation times 100, divided by the maximum command rate.

Hysteresis is the difference between outputs for increasing and decreasing input at that input for which the difference is maximum, measured after cycling through the input range and expressed as a per cent of one half of the output range.

4.18.2 Null voltage. With the gyro pickoff maintained at null as specified in 4.17.1 the output voltage of the torquer control winding shall be measured with a voltage measuring device as specified in 4.17.1.

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4.18.3 Reaction torque. The torquer reaction torque shall be determined by measuring the change in acceleration insensitive drift rate with the torquer reference winding energized and de-energized. The acceleration insensitive drift shall be measured using the method outlined in 4.27.

4.19 Temperature sensor tests.

4.19.1 Scale factor. The temperature sensor scale factor shall be determined by measuring the change in temperature sensor resistance for a 20 ±.1°F change in temperature in the range between 25°F below the operating temperature up to operating temperature. A wheatstone bridge shall be used for the resistance measurements. In order to reduce stabilization time in the ambient temperature, it is permissible to preheat the gyro to operating temperature. If a liquid bath is used, place the gyro in a plastic bag and secure the bag snugly around the gyro to force out excess air. The gyro shall have stabilized within ± .05°F at each ambient temperature before measurements of the sensor resistance may be used for determining the scale factor. The scale factor, expressed as ohms/°F., is the resistance change divided by twenty.

4.19.2 Induced voltage. With the gyro stabilized and excited as specified in 4.10.2 and 4.10.3 the temperature sensing winding shall be temporarily disconnected from the sensing circuit. The induced voltage in the temperature sensing winding shall be measured with a voltage measuring device as specified in 4.17.1.

4.19.3 Operating temperature. The purpose of this test is to determine the gyro temperature corresponding to the nominal value of operating temperature resistance specified in 3.10.3. The gyro ambient temperature shall be controlled to ±0.05°F of the nominal value of operating temperature specified in 3.10.6. The temperature sensing element resistance shall be measured using the procedure outline in 4.19.1. Determine the stabilized operating temperature of the gyro by using the scale factor as determined in 4.19.1 and the nominal value of operating temperature resistance specified in 3.10.3.

4.20 Heater power input. With the gyroscope at room temperature, the rated operating voltage as specified in 3.11.2 shall be applied separately to each of the heater windings in succession for a period of time necessary to determine the power input to each winding. Care should be taken not to over heat the gyro. Under these conditions, the power input to each winding shall be within the limits specified in 3.11.1.

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4.21 Gyro scale factor. Mount the gyro on a dividing head and test stand with the input reference axis parallel to the axis of rotation of the dividing head within ±0.5 degrees. The gyro shall be excited and stabilized as specified in 4.10.2 and 4.10.3. The gyro scale factor is measured with the gyro in open loop mode. A fixed bias current may be applied to the torquer to reduce the gyro drift. The gyroscope shall be rotated in both directions about the dividing head axis through 2 degrees ± 1 minute, 1 degree each side of the pickoff null position, at a rate of approximately 1 degree per second. The pickoff output voltage shall be measured to an accuracy of 1 per cent of the reading. The time for each reading shall not exceed one second, in order to minimize the effects of earth's rate and systematic drift rates. The change in pickoff voltage in the two directions shall be averaged and divided by the angle of rotation. Immediately after completing the gyro scale factor test the input angular freedom test shall be performed.

4.22 Input angular freedom. Input angular freedom shall be measured with the same test conditions and set up as specified in 4.21. Rotate the dividing head to the input limit and measure the pickoff output voltage. Reverse the direction of rotation and measure the pickoff output voltage at the opposite limit. The angular freedom shall be determined using the measured values of gyro scale factor and output limit voltage.

4.23 Gyro phasing. The gyro is phase-checked by testing spin motor, pickoff, and torquer phasing in sequence. All tests are done with the gyroscope excited and stabilized as specified in 4.10.2 and 4.10.3. Spin motor excitation phase sequence shall be as specified in the detail specification. Spin motor phasing is checked as follows: Orient the gyro OA horizontal with the SRA directed down and parallel to the dividing head spindle axis within 30 minutes of arc. Rotate the gyro about the SRA in a clockwise direction as viewed from above and observe the pickoff output voltage. For positive spin motor rotation, it will decrease and remain near null. Reversing thedirection of rotation should cause the gimbal to precess into a stop.

With proper spin motor phasing determined by the above procedure, pickoff phasing is checked as follows; Orient the gyro on the dividing head with the IA vertical within 30 minutes and directed down. Rotate the gyro about the IA in a clockwise direction as viewed from above until the gimbal precesses into a stop as indicated by the VTVM. Visually display a lissajous figure of the pickoff primary excitation voltage and the pickoff secondary voltage on an oscilloscope connected in the following manner: Secondary voltage to the horizontal amplifier. Primary voltage to the vertical amplifier. The line and return lines of the pickoff shall be as specified in the detail specifiecation.

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Oscilloscope should be phased such that an in phase signal produces a lissajous pattern which slants from lower left to upper right. Correct pickoff phasing exists if the secondary voltage leads the primary voltage by less than 90 degrees. A phase meter may be used in place of the oscilloscope for phase measurement.

Torquer phasing is checked after proper phasing has been established on the spin motor and the pickoff. No particular gyro orientation is required. Torque the gimbal into a stop by applying positive torque current as specified in the detail specification. Observe the lissajous figure of the pickoff primary and secondary voltage with the terminal connections used in pickoff phasing test. Correct torquer phasing is indicated if the pickoff produces a positive signal output.

4.24 Null repeatability. Mount the gyro OA vertical on a dividing head and test stand with the OA vertical and parallel to the dividing head axis of rotation. For the purpose of the test procedure that follows, a null position is defined as any turntable setting at which the precessional torque due to the earth's rotation opposes and balances the resultant of all other gimbal unbalance torques as evidenced by zero ±0.5 microampere of control winding current. The test procedure is as follows:

(1) Observe the test conditions as specified in 4.10. Connect the gyro into a closed rate loop as described in 4.11.

(2) Rotate the turntable clockwise (as seen from above) directing the IA eastward to find an east null position.

(3) Likewise, find and record a west null position for clockwise turntable rotation.

(4) Repeat step (2) and step (3), using counterclockwise turntable rotation.

(5) Average the CW and CCW east null positions determined in steps (2) and (3).

(6) Repeat step (5) using the west null position.

(7) Alternately rotate the turntable to the average null positions determined in steps (5) and (6) and record the total rebalance current. Use both (CW and CCW) directions of rotation to obtain four torque readings in each position.

(8) Note and record the greatest torque spread of any null position. To convert the spread to an equivalent drift rate in degrees per hour, it is necessary to multiply the current spread by a constant which is determined in the command rate scale factor test 4.18.1. A typical data tabulation for the null repeatability tests is shown in Table 2.

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TABLE II NULL REPEATABILITY TESTS

East Null Setting			West Null Setting		
CW	110 deg	5 min	CW	293 deg	10 min
CCW	110 deg	21 min	CCW	292 deg	44 min
	2/220 deg	26 min		2/585 deg	54 min
East Null	110 deg	13 min	West Null	292 deg	57 min

Rotation Top View	Rebalance Current (μa)	
	East Null	West Null
CW	+0.02	-0.01
CCW	+0.01	+0.02
CW	+0.03	+0.00
CCW	-0.01	+0.02
Max. Spread	0.04	0.03

Null Repeatability = (Max. Spread x Command Rate Scale Factor)

Command Rate Scale Factor = 2.02 deg/hr/μa

East Null Repeatability = 0.04 μa x 2.02 deg/hr /μa = 0.08 deg/hr

West Null Repeatability = 0.03 μa x 2.02 deg/hr/μa = 0.06 deg/hr

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4.25 Characteristic time. This test is used to determine the characteristic time of the gyro by measuring the phase angle between input and output when applying sinusoidally oscillating inputs to the gyroscope. Mount the gyro on a single-axis oscillating table with means of driving it sinusoidally at known variable frequency and means of reading phase angle between input and output. The gyro shall be positioned with the IA parallel to the table axis and shall be energized and stabilized as described in 4.10.2 and 4.10.3. Connect gyro pickoff and table angular position pickoff to a phase angle measuring device. Measure the frequency (f) at which the phase angle between gyro input (platform angular position) and gyro output is equal to a predetermined value (α). As an alternative, measure the phase angle (α) at a predetermined frequency (f). Compute the characteristic time (T) by use of the relationship $T = \frac{\tan\alpha}{2\pi f}$

4.26 Axis alignment. The alignment check procedure required that the turntable revolve at a constant speed and carry the gyro in the test fixture so that the axis of rotation is very accurately placed perpendicular to certain specified test fixture reference surfaces. The following procedure is used:

(1) Mount the gyroscope to the accuracy specified in 4.10.4 on the turntable using the test fixture, with the gyro OA parallel to the table axis. The gyro shall be energized and stabilized as specified in 4.10.2 and 4.10.3.

(2) With closed loop feedback to the torquer, adjust the loop with a dummy pickoff to the pickoff null and record the torquer reference and control windings current, and convert to the equivalent angular velocity by the use of the measured or design value o f command rate scale factor.

(3) Rotate the turntable approximately one revolution per second. While the speed is being timed adjust the feedback loop with the dummy pickoff to drive the gyro pickoff to null. Record the resulting feedback signal current to the torquer. Multiply by the excitation current and scale factor as in step (2). Correct this equivalent angular velocity by subtracting the initial reading taken in step (2).

(4) Divide the corrected equivalent angular velocity by the measured turntable angular velocity in radians per second. The result is the indicated misalignment of the gyro IA about the SRA in radians, with respect to the test arrangement.

(5) For the second phase of the alignment check place the gyro and fixture so that the gyro SRA is approximately parallel to the turntable axis, and repeat steps (2) through (4).

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The direction of rotation of the turntable should be positive with the spin reference axis according to the right-hand rule. The right-hand positive rotation avoids a possible instability that tends to drive the misaligned gyro gimbal by 180 degrees so as to line up the wheel rotation with the gyro case rotation. The result is the value of the indicated misalignment of the gyro input axis about the output axis.

4.27 Six position servo turntable test. The purpose of this test is to determine the systematic drift rate components consisting of acceleration insensitive drift rate and that portion of acceleration sensitive drift rate caused by mass unbalance. Test equipment includes base mounting fixtures, excitation supplies, single-axes servo driven turntable and associated servo amplifier and electronic timer. Observe the general operating requirements as specified in 4.10. Mount the gyro in the test fixture and connect the gyro pickoff to the servo which drives the single axis turntable. Excite and stabilize the gyro as described in 4.10.2 and 4.10.3. In this test setup the table rotation is equivalent to the gyro drift rate for that orientation. The gyro drift rate is measured by monitoring the table rotation rate through a predetermined fixed angle. The size of the angle, its repeatability, and precision of the time measurement must be compatible with the accuracy requirements of the drift test. The gyro is placed on the servo-table consecutively in each of six orientations. For each of these orientations five consecutive drift rate measurements are obtained, as described above. Each time, the table is returned to its original position by torquing the gyro and then allowed to drift again through the same precise angle for the next drift run.

MOUNTING POSITIONS:

Position 1	Position 2	Position 3
Input Axis up	Input Axis down	Input Axis south
Spin Axis east	Spin Axis west	Spin Axis west
Output Axis south	Output Axis south	Output Axis up
Position 4	Position 5	Position 6
Input Axis south	Input Axis south	Input Axis south
Spin Axis east	Spin Axis down	Spin Axis up
Output Axis down	Output Axis west	Output Axis east

For Position 1 and 2 the servo-table axis is vertical. For Positions 3 through 6 the table axis is horizontal in the north-south direction. This is achieved by tilting the table top toward the north. When the Input Axis vector is reversed with respect to the table axis, phasing of the gyro pickoff must be reversed, in order for the loop to servo properly. It will be noted that the gyro pickoff phasing for tests in Positions 2 through 6 is reversed from the phasing for test in

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Position 1. In Positions 3 and 4 the table drift angle must be centered around the Output Axis vertical position. In Positions 5 and 6 the table drift angle must be centered around the Output Axis horizontal position.

TEST RESULTS:

a. For each drift run, divide the fixed angle by the drift time, in order to obtain the drift rate. Compute the average drift rate for each of the six positions. Compute the systematic drift rate components contributed by acceleration sensitive drift rate and acceleration insensitive drift rate.

b. CW table rotation viewed from above = +.

c. CCW table rotation viewed from above = -.

d. Positions 1 and 2.

$$W'1 = W'EV + R12 - MUS$$

$$W'2 = W'EV - R12 - MUS$$

$$W'1 - W'2 = 2R12 \qquad R12 = \frac{W'1 - W'2}{2}$$

$$W'1 + W'2 = 2\,W'EV - 2MUS \qquad MUS = W'EV - \frac{W'1 + W'2}{2}$$

e. Position 3:

$$W'3 = W'EH - R3 \qquad R3 = W'EH - W'3$$

f. Position 4:

$$W'4 = W'EH - R4 \qquad R4 = W'EH - W'4$$

g. Position 5 and 6:

$$W'5 = W'EH - RS6 + MUI$$

$$W'6 = W'EH - RS6 - MUI$$

$$W'5 + W'6 = 2\,W'EH - 2\,RS6 \qquad RS6 = W'EH - \frac{W'5 + W'6}{2}$$

$$W'5 - W'6 = 2MUI \qquad MUI = \frac{W'5 - W'6}{2}$$

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h. In the above formulas the following symbols are used;

ω_i = Table angular rate in Position i.

ω_{ER} = Horizontal component of earth's rate = 15.041 cos (latitude angle) degrees per hour in Northern Hemisphere.

ω_{EV} = Vertical component of earth's rate = 15.041 sin (latitude angle) degrees per hour in Northern Hemisphere.

IA = Input Axis

OA = Output Axis

SA = Spin Axis

Ri = Acceleration insensitive drift rate in Position i.

MUI = Drift rate due to mass unbalance along IA.

MUS = Drift rate due to mass unbalance along SA.

Index numbers, i = 1,2-----,6 refer to the positions numbered as shown.

The acceleration insensitive drift rate is the maximum of the Ri values.

4.28 Six position rate loop test. The purpose of this test is to determine the systematic drift rate components consisting of acceleration insensitive drift rate and the acceleration sensitive drift rate caused by mass unbalance. Test equipment includes gyro mounting fixture, dividing head, excitation supplies, servo rate loop and torquer current readout. Observe the general operating requirements tstated in 4.10. Mount the gyro in the test fixture on the dividing head and connect the gyro pickoff to the input of the servo rate loop. A servo rate loop is described in 4.11. Excite and stabilize the gyro in accordance with 4.10.2 and 4.10.3. In this test the current in the torquer required to capture the gyro is proportional to its drift rate. The time used to obtain the average torquer current, the current measuring instrument repeatability and the command rate scale factor precision must be compatible with the accuracy requirements of the drift rate tests. Perform tests consecutively in each of the six orientations described.

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MOUNTING POSITIONS:

Position 1	Position 2	Position 3
Input Axis north	Input Axis south	Input axis up
Spin Axis east	Spin Axis west	Spin Axis west
Output Axis up	Output Axis up	Output Axis north

Position 4	Position 5	Position 6
Input Axis down	Input Axis east	Input Axis west
Spin Axis east	Spin Axis up	Spin Axis down
Output Axis north	Output Axis north	Output Axis north

NOTE: The output axis is mounted parallel to the table axis in all six positions.

TEST RESULTS:

a. Compute the drift rate for each of the six positions by multiplying the average torquing current by the command rate scale factor. Compute the systematic drift rate components contributed by acceleration sensitive and acceleration insensitive drift rates.

b. Sign convention. A rate loop equivalent drift rate is considered positive (+) if it tends to precess the gyro float CCW relative to the case when the gyro is viewed from +OA.

c. Analysis of Positions 1 and 2.

$$W1 = KJ1 = R12 + W_{EH}$$

$$W2 = KJ2 = R12 - W_{EH}$$

$$KJ1 - KJ2 = 2W_{EH} \qquad K = \frac{2W_{EH}}{(J1 - J2)}$$

$$KJ1 + KJ2 = 2R12 \qquad R12 = \frac{K}{2}(J1 + J2)$$

d. Analysis of Position3 and 4.

$$W3 = KJ3 = -MUS + R34 + W_{EV}$$

$$W4 = KJ4 = MUS + R34 - W_{EV} \qquad MUS = \frac{K}{2}(J4 - J3) + W_{EV}$$

$$KJ4 - KJ3 = 2MUS - 2W_{EV}$$

$$KJ4 + KJ3 = 2R34 \qquad K34 = \frac{K2(J4 + J3)}{2}$$

c. Analysis of Positions 5 and 6

$$W6 = KJ6 = -MUI + R56$$

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$W'S = KJ5 = MUI + R56$ $\qquad MUI = \frac{K(J5 - J6)}{2}$

$KJ5 - KJ6 = 2MUI$

$KJ5 + KJ6 = 2R56$ $\qquad R56 = \frac{K(J5 + J6)}{2}$

h. In the above analysis the following symbols are used:

$W'EH$ = Horizontal component of earth's rate = 15.041 cosine (latitude) deg/hr in Northern Hemisphere.

$W'EV$ = Vertical component of earth's rate = 15.041 sine (latitude) deg/hr in Northern Hemisphere.

$W'I$ = KJi = Equivalent drift rate in Position i due to torquer current

J = Torquer current.

K = Command rate scale factor

IA = Input Axis

OA = Output Axis

SA = Spin Axis

MUI = Drift rate due to mass unbalance along IA.

MUS = Drift rate due to mass unbalance along SA.

Ri = Acceleration insensitive drift rate in Position i

Index numbers, i = 1, 2,-------,6 refer to the positions numbered as shown.

The acceleration insensitive drift rate is the maximum of the Ri values.

4.29 Tumble test. The purpose of this test is to measure random drift rate, acceleration insensitive drift rate, and the acceleration sensitive drift rate components, mass unbalance and anisoelasticity. Test equipment consist of base, mounting fixture, excitation supplies, single axis turntable with precision drive, closed rate loop servo system and recorder. Observe the general operating conditions as specified in

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4.10. Connect the gyro in the closed rate loop as specified in 4.11. The gyro shall be mounted with its OA parallel to the turntable axis and parallel to the polar axis of the earth. Therefore, no component of earth's rate can be sensed by the gyro. The table is rotated at a rate of 300 degrees per hour for 4 complete turntable revolutions, and the restoring torquer current is recorded. The recording is synchronized with the table rotation so that the effect of gyro orientation with respect to gravity may be analyzed. The result is a trace approximating a sine wave. The restoring torquer current test results are then converted into drift rate, using the command rate sensitivity determined in 4.10.1. Perform a Fourier series expansion of the following form:

$$f(\beta) = C_0 + C_1 \sin \beta + C_2 \cos \beta + C_3 \sin 2\beta + C_4 \cos 2\beta$$

letting (β), the turntable angle, equal zero when the gyro's input axis is vertical. The following information can then be obtained:

1. Acceleration - insensitive drift rate: This corresponds to the bias or d - c level of the trace (Fourier coefficient C_0)

2. Acceleration - sensitive drift rate: This is equal to the amplitude of the fundamental of the recorded trace $\sqrt{C_1^2 + C_2^2}$ divided by the cosine of the latitude angle.

3. Drift rate due to anisoelasticity: This is equal to the amplitude of the second harmonic of the trace $\sqrt{C_3^2 + C_4^2}$ divided by the cosine of the latitude angle squared. This effect, however, may be obscured by the gyro's random drift rate.

4. Random drift rate: This can be computed by either of the alternate methods described below:

a. For any one cycle at each 10 degree increment of table rotation (β), subtract the corresponding Fourier coefficients from the raw data. The standard deviation of these 37 residual values is the random drift rate.

b. For successive cycles determine the difference in total drift rate between any two successive cycles at each corresponding 10 degree increment of table rotation (β). The standard deviation of these 37 differences is the random drift rate.

4.30 Servo turntable random drift rate. The gyroscope shall be mounted on a turntable as specified in 4.11 with either of the two following orientations; OA horizontal, and IA vertical, or OA vertical and IA horizontal in a North-South direction. Excite and stabilize the gyro as specified in 4.10.2 and 4.10.3. The gyro pickoff signal is amplified and used to operate a servo-motor that drives the turntable about the input axis of

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or maximum deviation of torquer current, or both, shall be as specified in any portion of the cycle. The cycle shall be repeated as specified in the detail specification.

4.34.5 Spin motor excitation variation. Unless otherwise specified, wheel voltage shall be increased by 10 per cent. Maximum permissible change in drift or unbalance terms shall be prescribed by the detail specification.

4.34.6 Spin motor excitation frequency variation. Unless otherwise specified, wheel frequency shall be increased by one percent. Maximum permissible change in drift or unbalance terms shall be prescribed by the detail specification.

4.34.7 Pickoff excitation variation. Unless otherwise specified, the pickoff excitation shall be increased by 10 per cent. Maximum permissible change in drift or unbalance terms shall be prescribed by the detail specification.

4.34.8 Torquer excitation variation. Unless otherwise specified, the torquer excitation shall be increased by 10 per cent. Maximum permissible change in drift or unbalance terms shall be perscribed by the detail specification.

4.35 Vibration test. The gyro shall be rigidly mounted on the vibration equipment utilizing normal mounting surfaces. The gyro shall then be subjected to sinusoidal vibration of 0.06 inches peak-to-peak amplitude, or a peak acceleration of fifteen (15) gravitational units, which ever is less, over the range of 10 to 2000 cps in each of three mutually perpendicular axes for a period of four (4) hours. The logrithmic frequency sweep from 10 to 2000 to 10 cps shall require 30 minutes to complete. During vibration the gyro shall be excited and stabilized as specified in 4.10.2 and 4.10.3 in a servo rate loop as described in 4.11.

4.36 Shock test. The gyro shall be subjected to 30 impacts of 50 g's peak acceleration and 11 ± 1 milliseconds duration. The shock pulse may be produced by any method capable of producing a pulse approximating a half sine with less than 10 per cent peak pulse distortion at any point on the pulse. The gyro shall be rigidly mounted to the shock test equipment utilizing normal mounting surfaces. During the test the gyro shall not be excited. The gyro shall be subjected to five impacts in each direction along the three mutually perpendicular axes of the gyro (30 impacts).

Following the test, the gyro shall meet the requirements of 3.25.

4.37 Altitude test. After placing the gyro in a vacuum chamber, reduce the pressure to simulate an altitude of 100,000 feet (approximately 8mm of Hg). While still subjected to this environment the gyro shall be tested according to the requirements of 3.26.

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4.38 Moisture resistance. Gyros shall be tested to the requirements of MIL-STD-202, Method 106. The gyro shall be initially unexcited. At the end of each 16 hour period in the cycle (completion of step 6) and after the humidity is controlled at 90-95 per cent, the gyro shall be excited with rated standard test voltages and remain so for a period of five cycles. After the ten cycles have been completed, the gyro shall remain unexcited for a period of at least 24 hours at the standard ambient condition of 4.10.1 and subsequently meet the requirements of 3.27.

4.39 Salt atmosphere resistance. Gyros shall be tested in accordance with MIL-STD-202, Test Method 101A, Test Condition B. The gyro shall be placed in the test chamber with the output axis horizontal and in such a manner as to permit circulation of the mist about the gyro. All mounting or suspension parts which are in direct contact with the gyro during this test shall be materials that will not cause corrosion. After subjection to this test, the gyro shall meet the requirments of 3.28.

4.40 Ambient temperature test. The gyro shall be placed in a test chamber and the chamber temperature reduced to the low temperature extreme specified in the detail specifications. This temperature shall be maintained for at least 24 hours. The gyro shall then be placed in a standard ambient 72 ± 5°F. for a period of 4 hours before testing to the requirements of 3.29.

4.41 Radio noise. The gyroscope shall be excited and stabilized as specified in 4.10.2 and 4.10.3. Radio interference radiated into space and conducted into the line shall be measured in accordance with the methods given in specification MIL-I-16910 and shall be as specified in 3.30.

4.42 Operating life. The gyro shall be excited and stabilized as specified in 4.10.2 and 4.10.3 in a servo rate loop as described in 4.11. At periodic intervals of 400 hours, measure the performance characteristics specified in 3.31. The unit shall meet all the requirements specified by 3.31 each time the performance characteristics are measured. Failure of the gyro due to a deviation from the performance specification shall terminate the test. Operational life shall be considered as the total hours accumulated at the end of the last performance check in which all characteristcs were within specification.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging.

5.1.1 Level A

5.1.1.1 Cleaning, drying, and preservation. Not applicable.

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5.1.1.f Unit packaging. Unit packaging of the gyroscope shall be in accordance with Method IIB of Specification MIL-P-116. The cushioning material shall have a PH value between 6.0 and 8.0 and shall not produce corrosive or deleterious fumes or vapors which could act on or attack the gyros.

5.1.2 Level C.

5.1.2.1 Cleaning, drying, preservation and packaging. Unless otherwise specified in the contract or order, cleaning, drying preservation and packaging shall be in accordance with 5.1.1.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. Gyroscopes packaged as described in 5.1.1.2 shall be packed in overseas type shipping containers meeting requirements specified in PPP-B-621. The gross weight of each pack shall be limited to approximately 70 pounds.

5.2.2 Level B. Unless otherwise specified, gyros packaged in accordance with 5.1.1 shall be packed in domestic type shipping containers meeting requirements specified in PPP-B-621. The gross weight of each pack shall be limited to approximately 70 pounds.

5.23 Level C. Gyros packaged in accordance with 5.1.1 shall be packed in exterior domestic type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification rules or regulations of common carriers as applicable to the mode of transportation.

5.3 Marking. Unless otherwise directed, interior and exterior containers shall be marked in accordance with MIL-STD-129. The shipping nomenclature shall be as follows "Gyro Rate Integrating."

5.4 Tests.

5.4.1 Rough handling tests. A preproduction pack of gyroscopes , packaged and packed in accordance with 5.1 and 5.2 shall be subjected to appropriate rough handling tests as specified in paragraph 4.4.6 of MIL-P-116 and upon completion, the gyros shall be tested and inspected against operational malfunction in accordance with 4.1. Preproduction tests shall be repeated when changes in packaging and packing materials, processes, or design are made.

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6. NOTES.

6.1 Intended use. Gyroscopes covered by this specification are intended for use in applications requiring space stabilization or measurements of angular rate of change from a fixed reference. This gyroscope design has successfully endured the environmental extremes outlined herein. However, because of the precision nature of this device severe exposure, particularly to low temperature, is to be avoided for improved reliability and performance. To avoid buildup of excessive internal pressure, do not expose or operate the gyroscope at temperatures exceeding specified limits. For best drift stability and spin motor life, the spin motor should not be energized when the gyroscope is at lower than operating temperature (3.10.6).

6.2 Ordering data. Procurement documents specify the following:

(a) Title, number, and date of this specification.

(b) Level of shipment required (see 5.1 and 5.2).

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